

# Cristhian Núñez Ramos

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## Curriculum Vitae

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### 1. Education

- 2022–2025 **Ph.D. in Mathematics**, *Pontificia Universidad Católica de Chile*, Santiago-Chile.  
Title of Thesis: Symplectic Hamiltonian Discontinuous Galerkin Methods for Shallow Water Equations. Advisor: Manuel A. Sánchez
- 2021–2022 **Master in Mathematics**, *Pontificia Universidad Católica de Chile*, Santiago-Chile.
- 2015–2018 **Master in Optimization**, *Escuela Politécnica Nacional*, Quito-Ecuador.  
Title of Thesis: Numerical Simulation of Free Boundary Bingham Fluids. Advisor: Pedro Merino
- 2009–2014 **Bachelor in Pure Mathematics**, *Escuela Politécnica Nacional*, Quito-Ecuador.  
Title of Thesis: A Proper Orthogonal Decomposition for the Fisher Equation. Advisor: Pedro Merino

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### 2. Areas of interest

My research focuses on Numerical Analysis of Partial Differential Equations, particularly Finite Element Methods. Specifically, I am interested in Discontinuous Galerkin Methods, Hybridizable Discontinuous Galerkin Methods, and the development of novel numerical methods for Hamiltonian systems that preserve key physical invariants in long-term simulations. Additionally, my interests extend to the optimal control of PDEs, with applications in mathematical modeling, computational physics, and Engineering problems requiring accurate and robust simulations.

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### 3. Research Experience

- 2024 **Researcher (Internship)**, *School of Mathematics, University of Minnesota*, Minneapolis, United States.  
Developed numerical methods for solving the vector Laplacian equation.
- 2021–2025 **Ph.D. Student**, *Facultad de Matemáticas, Pontificia Universidad Católica de Chile*, Santiago, Chile.  
Developed novel numerical methods for shallow water equations.
- 2015–2021 **Researcher**, *Centro de Modelización Matemática, Escuela Politécnica Nacional*, Quito, Ecuador.  
Developed and analyzed mathematical models and algorithms for viscoplastic fluid simulations.
- 2015–2016 **Researcher**, *Facultad de Ciencias – Instituto Geofísico del Ecuador*, Quito, Ecuador.  
Designed algorithms for image classification and modeling of volcanic lava and viscoplastic flows.

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## 4. Academic Appointments

25/08/2025 **Postdoctoral Associate**, *University of Minnesota*, Minneapolis, USA.

**Menthor:** Bernardo Cockburn.

We are developing post-processing techniques for elliptic problems that systematically enhance the quality of approximations produced by Galerkin methods by exploiting hidden properties of the governing equations.

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## 5. Published Papers (Refereed Journals)

1. Merino, P. M., & Núñez, C. A. (2024). Numerical simulation of free boundary biviscous fluids. *Bulletin of Computational Applied Mathematics*, 12(2), 163–188.
2. Núñez, C., & Sánchez, M. A. (2025). Symplectic Hamiltonian hybridizable discontinuous Galerkin methods for linearized shallow water equations. *Computer Methods in Applied Mechanics and Engineering*, 447, 118383. <https://doi.org/10.1016/j.cma.2025.118383>

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## 6. Manuscripts Submitted

1. *HDG methods for the two-dimensional vector Laplacian* with Bernardo Cockburn and Manuel Sánchez.

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## 7. Manuscripts in Preparation

1. *Turbo Post-Processing for Mixed Methods for One-dimensional Elliptic Problems* with Bernardo Cockburn.
2. *Symplectic Hamilton–HDG Methods for Nonlinear Shallow Water Equations* with Manuel Sánchez.
3. *POD Techniques for an Optimal Control Problem of Fisher equation* with Pedro Merino.
4. *Regularization-based approach for solving the shallow water equations with Bingham viscoplastic flows using discontinuous Galerkin methods*, with Sofía López y Felipe Fernández.

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## 8. Other Publications

1. Núñez Ramos, C. A., & Villacís Tulcán, N. E. (2025). Simulation of an optimal control of the epidemiological reservoir–people model. *Matemática (ESPOL–FCNM Journal)*, 23(2).
2. Navarrete Vaca, L., & Núñez Ramos, Cristhian (2025). Control and communication in dialogue: controllability, observability, and bang–bang dynamics. *Selecciones Matemáticas*, 12(2), 365–374. <https://doi.org/10.17268/sel.mat.2025.02.08>
3. Núñez Ramos, C. A., & Pizanan Cárdenas, C. H. (2026). Optimal control of the Fisher equation: a comparison of steepest descent and quasi-Newton methods. *Matemática (ESPOL–FCNM Journal)*, 24(1).

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## 9. Presentations

August 11–15, 2025 **XII International Congress on Applied and Computational Mathematics (XII CIMAC)**, *Universidad Nacional del Altiplano*, Puno, Perú.

Oral presentation: “HDG Methods for the two-dimensional Vector Laplacian Equation”

March 25–28, 2025 **1st Heidelberg–Chile Workshop on Scientific Computing**, Santiago, Chile.

Poster presentation: “Hybridizable Discontinuous Galerkin Methods for the Vector Laplacian Equation”

- March 17–20, 2025 **IACM Computational Fluids Conference (CFC)**, Santiago, Chile.  
Oral presentation: “Symplectic Hamiltonian HDG Methods for Linearized Shallow Water Equations”
- June 12–14, 2024 **Conferencias Colombianas de Matemáticas Aplicadas e Industriales (MAPI3)**, Bucaramanga, Colombia.  
Oral presentation: “Symplectic Hamiltonian HDG Methods for Linearized Shallow Water Equations”
- April 24–26, 2024 **XXXVI Jornada de Matemática de la Zona Sur**, *Universidad Católica de Temuco*, Temuco, Chile.  
Oral presentation: “Symplectic Hamiltonian HDG Methods for Shallow Water Equations”
- January 15–19, 2024 **Workshop on Numerical Analysis of Partial Differential Equations (WON-APDE)**, *Universidad de Concepción*, Concepción, Chile.  
Oral presentation: “Symplectic Hamiltonian HDG Methods for Linearized Shallow Water Equations”
- October 10–13, 2023 **XLIX Semana de la Matemática**, *Pontificia Universidad Católica de Valparaíso*, Valparaíso, Chile.  
Oral presentation: “Métodos Hamiltonianos Simplécticos HDG para las ecuaciones de aguas poco profundas”
- May 29–31, 2023 **Encuentro Latinoamericano de Estudiantes de Doctorado en Modelamiento, Ingeniería y Ciencias (EMIC)**, *Universidad de Chile*, Santiago, Chile.  
Oral presentation: “Symplectic Hamiltonian HDG Methods for Shallow Water Equations”
- November, 14–19, 2022 **XVII Encuentro de Matemática y sus Aplicaciones**, *Escuela Politécnica Nacional*, Quito, Ecuador.  
Oral presentation: “Numerical Simulation of Free Boundary Biviscous Fluids”
- September 3–7, 2018 **VI Latin American Workshop on Optimization and Control**, *Escuela Politécnica Nacional*, Quito, Ecuador.  
Oral presentation: “Numerical Simulation of Free Boundary Bingham Fluids”
- November, 11th 2015 **III Concurso de Reconocimiento a la Investigación Universitaria Estudiantil – Galardones Nacionales**, Quito, Ecuador.  
Oral presentation: “Reducción de un Problema de Control Óptimo usando técnicas POD”

## 10. Academic Service

- 2025 Workshop on Applied Mathematics, Pontificia Universidad Católica de Chile, Santiago, Chile. Organizer. Responsible for scientific coordination, invited speakers, program design, and workshop logistics. <https://appliedmathuc.github.io/workshop-applied-math-2025/>

## 11. Workshops attended

- 2023 XXXV Jornada de Matemática de la Zona Sur, Universidad de Concepción, Chile
- 2023 2nd Chile-Japan Workshop on Mathematical Physics and PDE, Universidad de Santiago de Chile.
- 2022 Workshop on Scientific Machine Learning: Mathematics and Applications, Santiago, Chile.

2022 Workshop on Minimum Residual and Least-Squares Finite Element Methods, Santiago, Chile.

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## 12. Teaching Experience

10/01/2018– **Adjunct Professor**, *Escuela Politécnica Nacional*, Quito, Ecuador.

04/05/2021 Courses taught:

- Numerical Analysis for PDEs (2018-II, two sections)
- Numerical Analysis I (2018-II, two sections)
- Numerical Analysis II (2018-II, two sections)
- Linear Algebra (2018- I-II, two sections)
- Calculus of variations (2019- I-II, two sections)
- Modelling of PDEs (2019- I-II, two sections)
- Numerical Analysis II (2019-I-II, two sections)
- Biomathematics (2020-I-II, one sections)
- Numerical Analysis for PDEs (2020-I-II, one section)
- Non-Linear Analysis (2021-I, two sections)
- Numerical Analysis for PDEs (2021-I, one section)
- Biomathematics (2021-I, one section)

10/16/2017– **Adjunct Professor**, *Full Time*, *Escuela Politécnica Nacional*, Quito-Ecuador.

02/28/2019 Courses taught:

- Numerical Analysis I (2017-II, two sections)
- Numerical Analysis II (2017-II, two sections)
- Algebra (2017-II, one section)
- Calculus II (2018-I, two sections)
- Optimization I (2018-I, two sections)
- Differential Equations (2018-I, two sections)
- Calculus II (2018-II, two sections)
- Optimization I (2018-II, two sections)
- Optimization I (2018-II, two sections)

01/03/2017– **Adjunct Professor**, *Full Time*, *Escuela Politécnica Nacional*, Quito-Ecuador.

09/11/2017 Courses taught:

- Numerical Analysis II (2017-I, two sections)
- Calculus I (2017-II, two sections)
- Calculus II (2017-I, two sections)
- Complex Calculus (2017-I, four sections)
- Analysis I (2017-I, two sections)

03/05/2025– **Teaching Assistant**, *Pontificia Universidad Católica de Chile*, Santiago, Chile.

07/14/2025 Teaching assistance during doctoral studies, including support in lectures and grading in the course of Fundamentos Matemáticos para la Inteligencia Artificial.

09/01/2009– **High School Teacher**, *Plan Internacional Programa Bolívar*, Guaranda, Ecuador.

04/30/2012 Teaching Mathematics and Geometry at secondary education level.

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## 13. Professional and Consulting Experience

2017–2018 **Sampling Analyst (Consulting Services)**, *Instituto Nacional de Evaluación Educativa*, Quito, Ecuador.

Development of data-driven methods to support educational evaluation and the design of analytical tools for measuring academic progress.

09/01/2014– **Sampling Assistant**, *Instituto Nacional de Estadísticas y Censos*, Quito, Ecuador.  
03/31/2015 Design and implementation of statistical sampling methodologies for unemployment studies in Ecuador.

## 14. Early Career Experience

02/03/2014– **Undergraduate Intern**, *Escuela Politécnica Nacional*, Quito, Ecuador.  
06/30/2014 Teaching assistance in Mathematics and Geometry and academic support for undergraduate students.

## 14. Memberships and Awards

- Postdoctoral Associate. School of Mathematics, University of Minnesota (2025 –)
- VRI–UC Research Scholarship, Pontificia Universidad Católica de Chile (2025).
- Gubernamental ANID fellowship for doctoral studies in Mathematics. Chile (2021-2024)
- Associate Researcher of Centro de Modelización Matemática - Escuela Politécnica Nacional - Ecuador (2015-2021)
- Member of Sociedad Ecuatoriana de Matemáticas - Ecuador (2015–)
- PIGR-22-03 project grant, awarded by the Vicerectorado de Investigación, Escuela Politécnica Nacional, for research in Mathematical Optimization. (2016-2018)
- EPN fellowship for master studies in Optimization. Escuela Politécnica Nacional. Ecuador (2015-2017)

## 15. Skills

Comunicación **Languages**, *Spanish: Native*, English: Professional working proficiency.

Programming **Technical Skills**, Python, Matlab, SPSS, Stata, R/RStudio, C/C++, SQL, SML, Languages Scheme, SQL, LaTeX.